

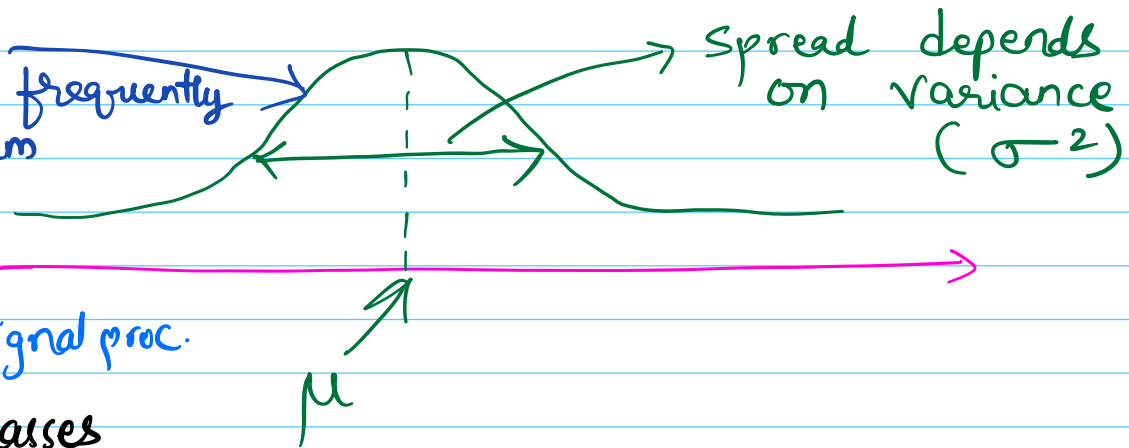
Lec: 12

Gaussian Random Variable

Gaussian
arises very frequently
in Linear System
Analysis

ex: Noise in
Communications / signal proc.

ML: different classes
are Gaussian.



$$\text{PDF } f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$-\infty < x < \infty$$

$$E\{x\} = \mu \longrightarrow \text{mean}$$

$$E\{(x-\mu)^2\} = \sigma^2 \longrightarrow \text{variance}$$

Gaussian Random Vector

Multi variate Gaussian

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{Jointly Gaussian}$$

$$E\{\bar{x}\} = \bar{\mu}$$

$$E\{(\bar{x} - \bar{\mu})(\bar{x} - \bar{\mu})^T\} = R$$



Covariance matrix

PDF of multivariate
Gaussian

$$f_X(\bar{x}) = \frac{1}{\sqrt{(\sqrt{2\pi})^n |R|}} e^{-\frac{1}{2}(\bar{x} - \bar{\mu})^T R^{-1}(\bar{x} - \bar{\mu})}$$

determinant

if $\bar{\mu} = 0$

$$E\{\bar{x}\} = 0$$

$$E\{\bar{x} \bar{x}^T\} = R$$

$$E\{\bar{x} \bar{x}^T\} = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ r_{21} & r_{22} & \dots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \dots & r_{nn} \end{bmatrix}$$

$$r_{ii} = E\{x_i^2\}$$

diagonal
entries

$$r_{ij} = r_{ji} = E\{x_i x_j\}$$

Correlation

off-diagonal
entries

Special case: $\bar{\mu} = 0$

$$E\{\bar{x} \bar{x}^T\} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma_n^2 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_R$

R is Diagonal

$$E\{x_i x_j\} = 0 \quad i \neq j$$

$\Rightarrow x_i, x_j$ are Uncorrelated

Gaussian
 $\Rightarrow$

x_i, x_j are INDEPENDENT
only for Gaussian.

Further, for

$$R = E\{\bar{x} \bar{x}^T\} = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma^2 \end{bmatrix}$$

$$= \sigma^2 \mathbf{I}$$

$$\propto \mathbf{I}$$

$$\Rightarrow E\{x_i^2\} = \sigma^2$$

$$E\{x_i x_j\} = 0 \quad i \neq j$$

$\Rightarrow x_i$ are INDEPENDENT

& x_i have identical mean/variance

$$\text{mean} = 0$$

$$\text{variance} = \sigma^2$$

Therefore, $x_1, x_2, \dots, x_n$

'i.i.d.'

$\rightarrow$ independent & identically
distributed
Gaussian RVs.

! My name is Keerthana

